

Jay Gandhi

Robotics & Embedded Systems Engineer (Undergraduate)

+91-7821037997 | jaygandhi1314@gmail.com | linkedin.com/in/jay-gandhi57 | github.com/Jay86935

EDUCATION

JSPM Rajarshi Shahu College of Engineering

B.Tech – Automation & Robotics Engineering (CGPA: 8.29 / 10)

Pune, India

2023 – 2027

RESEARCH EXPERIENCE

Research Intern – Robotics & Autonomous Systems

Indian Institute of Science (IISc), Bengaluru

Feb 2026 – Present

Bengaluru, India

- Currently contributing to an advanced robotics research project focusing on autonomous perception architectures and embedded control for dynamic environments.
- Learning and implementing ROS (Robot Operating System) frameworks to bridge high-level path planning algorithms with edge-computing hardware.
- Deploying real-time vision pipelines and autonomous navigation models onto embedded Linux systems, including Nvidia Jetson Nano and Raspberry Pi.
- Developing low-level hardware interfaces utilizing STM32 microcontrollers for precise sensor data acquisition (IMU, encoders) and motor actuation.
- Engineering custom sensor fusion nodes to optimize the perception-to-actuation latency loop, significantly reducing localization drift during autonomous traversal.

EMBEDDED & ROBOTICS SYSTEMS EXPERIENCE

Founder & Avionics Lead

Team Dhruvishaya – Rocketry & Space Research

Feb 2025 – Present

- Founded collegiate rocketry team and designed full avionics stack – power, sensing, telemetry, and ignition – for an experimental solid-propellant rocket platform.
- Built ignition and motor-control electronics for static fire testing; implemented hardware interlocks for launch safety.
- Developed real-time sensor monitoring system (pressure, temperature, IMU) with on-board data logging and wireless ground-station telemetry.
- Led end-to-end embedded system integration from schematic to firmware to field validation.

Electronics Lead – Janatics Automation Skill Challenge (JASC)

Industrial Automation Project – Semi-finalist | GitHub

Sep 2025 – Nov 2025

- Designed automated industrial waste-sorting system classifying materials at ~95% accuracy using inductive, capacitive, and IR sensors.
- Built PLC-Arduino hybrid control architecture (Siemens TIA Portal + Arduino Mega) enabling real-time sorting at ~30 items/min throughput.
- Led full system integration, wiring, and validation under industrial task constraints; reached semi-finals out of 200+ teams nationally.

PROJECTS

AgroBot – Autonomous Agricultural Monitoring Robot

GitHub

Mar 2026 – Apr 2026

- Built differential-drive robot for autonomous field monitoring – implemented line-following navigation using OpenCV-based lane detection on a Raspberry Pi camera stream.
- Integrated multi-sensor input (ultrasonic, IR, IMU) for real-time obstacle avoidance and path correction.
- Developed Wi-Fi telemetry module (ESP32) streaming live sensor data and camera feed to a web dashboard for remote monitoring.
- Demonstrated reliable autonomous traversal across a 5m × 3m test grid with <2% deviation from target path.

GitHub

- Designed 1 km-range RF control system using nRF24L01+ modules with custom packet protocol and adaptive retry logic.
- Implemented failsafe motor-disarm logic triggered on signal timeout (<50 ms response), validated in field tests.
- Achieved 99% packet delivery rate at 800 m LOS; documented for reuse as platform communication layer in rocketry ground-station.

TECHNICAL SKILLS

Languages: C, C++, Embedded C, Python

Robotics & Perception: ROS/ROS2, OpenCV, Sensor Fusion, SLAM (exposure), Stereo Vision, Motor Control

Hardware & Embedded: STM32, ESP32, ESP8266, Arduino, Jetson Nano (Nvidia), nRF24L01, Raspberry Pi

Automation: PLC Programming (Siemens TIA Portal), Sysmac Studio, Factory I/O

Tools: MATLAB, KiCad, Fusion 360, LTspice

CERTIFICATIONS

Industry 4.0 – C4i4 Lab, Pune

Industrial Automation Workshop – OMRON

ACHIEVEMENTS & RECOGNITION

Semi-finalist – Janatics Automation Skill Challenge (JASC), national-level industrial automation competition

Research Internship – Indian Institute of Science (IISc), Bengaluru (highly selective, 3rd-year undergraduate)

Collaboration exposure – Purdue EPICS Program (international student engineering partnership)